Electronic Parameters

1 Overview of the base electrical interface

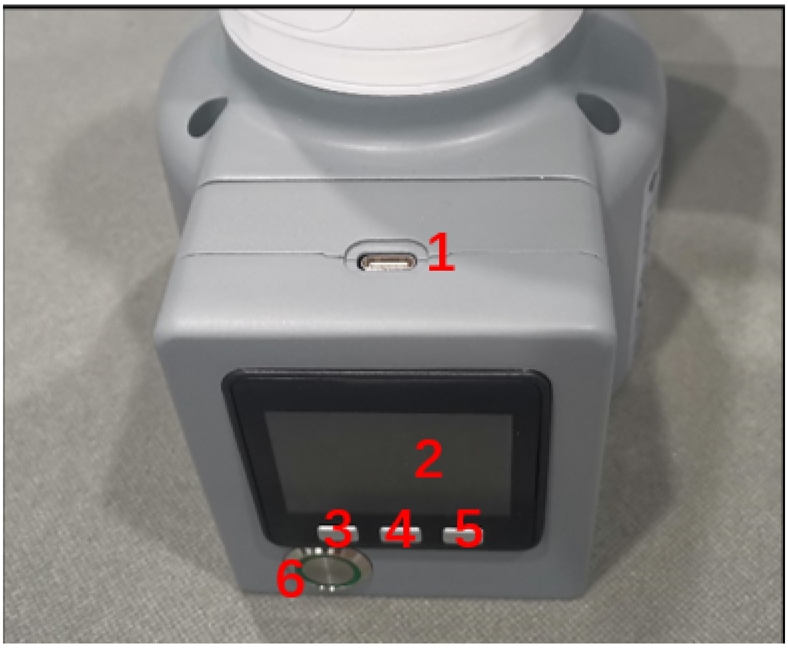

Figure 1 Front view of the base

Figure 2 Left side of the base

Figure 3 Right side of the base

Serial number	Interface Name	definition	Function	Remark
1	Type C	Communication interface	Communicate with PC	Development and use
2	Screen	show	Use with buttons	
3	button	Button A	Use with display	
4		Button B		
5		Button C		
6	switch	Power switch	Control input power on and off	With light (light turns on when powered on)
7	DC/IO Interface	GND	GND	
		IN6	Digital input signal 1~6	Input only in NPN mode
		IN5		
		IN4		
		IN3		
		IN2		
		IN1		
		24V	DC24V	

Serial number	Interface Name	definition	Function	Remark
8	Type C	Communication interface	Communicate with PC	Development and use
9	Power DC input interface	DC24V input	DC24V input	
10	DC/IO Interface	24V	DC24V	
		OUT1	Digital output signal 1~6	Output only in PNP mode
		OUT2		
		OUT3		
		OUT4		
		OUT5		
		OUT6		
		GND	GND	
11	Emergency stop interface	STOP	Emergency stop circuit interface	

1.1 Type C: The Type C interface is used to connect and communicate with a personal computer and can be used by developers.

1.2 Screen: The screen is used to display the communication status of myCobot and calibrate the robot to move to the starting point through a 2-inch IPS screen.

1.3 Button A, Button B and Button C are used to operate the screen in a coordinated manner.

1.4 Power switch: The power switch is used to control the main power input. If it is turned off, the controller will also be powered off.

1.5 24V output: Built-in DC24V, available for users.

1.6 Digital Input/Output: Includes 6 digital input signals and 6 digital output signals, which are used to interact with other devices and together with other devices constitute an important part of the automation system.

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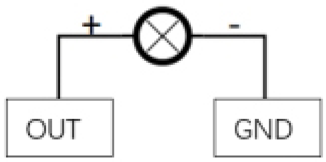
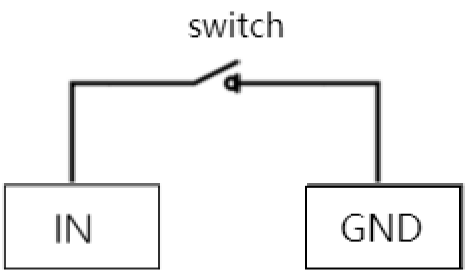
It should be noted that the output signal is in PNP form and the input signal is in NPN form. The following is the external wiring diagram:



Example of Output Signal Wiring



Example of Input Signal Wiring

1.7 Power DC input interface: Use KPPX-4P R7BFDC power socket. The 24V 9.2A DC power adapter provided by the manufacturer can also be used to power myCobot320.

1.8 The emergency stop circuit terminal is connected to the emergency stop button box and can be used to control the emergency stop of the robot.

> **Note** : The emergency stop switch must be connected when using the robot, and make sure the emergency stop switch circuit is always connected.

2 Electrical interface at the end of the robot arm

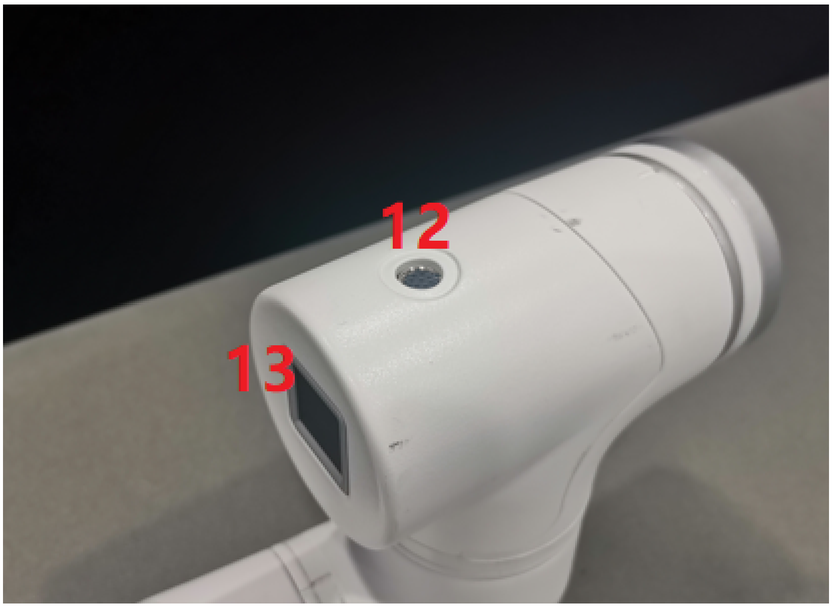


Figure 4 Side view of the end of the robotic arm

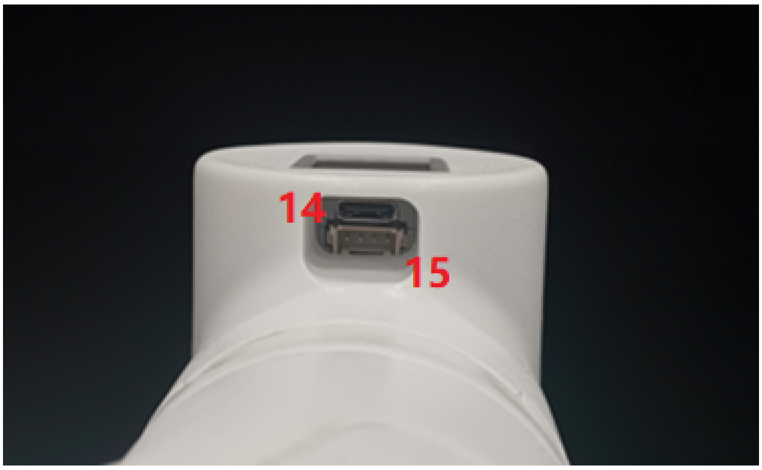
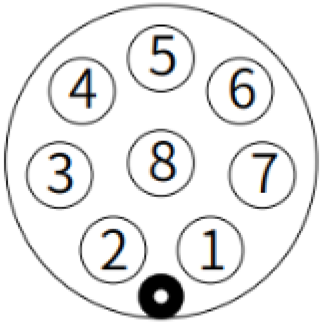


Figure 5 Side view of the end of the robotic arm

Serial number	Interface Name	definition	Function	Remark
12	M8 aviation socket	Terminal tool IO interface	Terminal tool IO interface	
13	Atom	LED + Button	State View/Drag Tutorial	
14	Type C	Communication interface	Communicate with PC	To update the Atom firmware use
15	Grove			Developer Use

2.1 Terminal tool IO interface:

This is a tool input/output diagram, the myCobot 320 robot provides one input and two outputs.



FRONT VIEW

Tool I/O Diagram

Built-in I/O power supply, voltage 24V, single output maximum 0.2A. Does not support external power supply.

The definition of each tool I/O port is shown in the following table. It should be noted that the input and output of the tool I/O are both PNP type, and the wiring mode is the same as the bottom output interface.

Serial number	Signal	illustrate	Matching M8 wire color
1	GND	DC24V negative	白
2	OUT1	Tool output interface 1	褐
3	OUT2	Tool output interface 2	绿
4	485A	Reserved, undeveloped	黄
5	24V	DC24V positive	灰
6	IN1	Tool input interface 1	粉
7	IN2	Unavailable	蓝
8	485B	Reserved, undeveloped	紫

2.2 Atom: Atom is used for 5X5 RGB LED to display the robot arm status and button functions (used when the robot performs drag teaching)